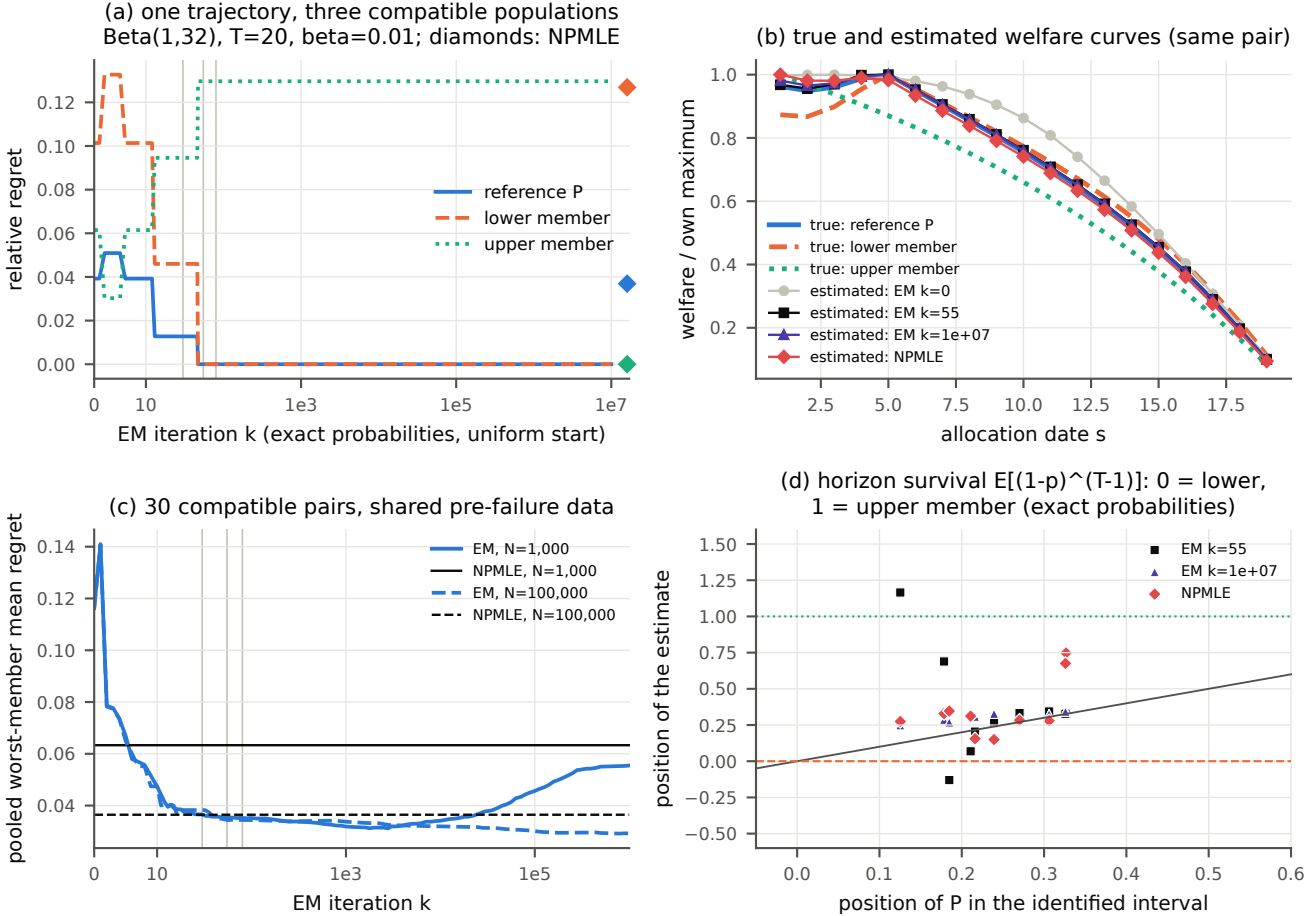
(a) one trajectory, three compatible populations
Beta(1,32), T=20, beta=0.01; diamonds: NPMLE
relative regret
0.12
0.10
0.08
0.06
0.04
0.02
0.00
reference P
lower member
upper member
0
10
1e3
1e5
1e7
EM iteration k (exact probabilities, uniform start)
(b) true and estimated welfare curves (same pair)
welfare / own maximum
1.0
0.8
0.6
0.4
0.2
true: reference P
true: lower member
true: upper member
estimated: EM k=0
estimated: EM k=55
estimated: EM k=1e+07
estimated: NPMLE
2.5
5.0
7.5
10.0
12.5
15.0
17.5
allocation date s
(c) 30 compatible pairs, shared pre-failure data
pooled worst-member mean regret
0.14
0.12
0.10
0.08
0.06
0.04
EM, N=1,000
NPMLE, N=1,000
EM, N=100,000
NPMLE, N=100,000
0
10
1e3
1e5
EM iteration k
(d) horizon survival E[(1-p)^(T-1)]: 0 = lower,
1 = upper member (exact probabilities)
position of the estimate
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
EM k=55
EM k=1e+07
NPMLE
0.0
0.1
0.2
0.3
0.4
0.5
0.6
position of P in the identified interval